CrisisRelief AI: Disaster Response Message Classification and Triage System

Domain: Mission / Crisis / Disaster Response

Executive Summary

During natural disasters such as earthquakes, floods, and hurricanes, emergency response agencies receive thousands of unstructured text messages, social media posts, and SMS feeds. Rapidly triaging these short communications to isolate actionable emergency requests from general background commentary is critical for allocating life-saving resources.

CrisisRelief AI is an end-to-end Machine Learning pipeline and web application designed to classify short communications from disaster zones into disaster-related vs. non-disaster-related priorities using Natural Language Processing (NLP) feature extraction and explainable predictive models.

Dataset Overview and Academic Citations

- **Dataset Source**: Robert Munro - Disaster Response Messages Repository
- **Dataset Size**: 25,000+ short messages collected during major historical disaster events:
  - 2010 Haiti Earthquake (SMS via Mission 4636)
  - 2010 Pakistan Floods (SMS via PakReport)
  - 2012 Hurricane Sandy (USA online message boards)
  - Global disaster news headlines
- **Category Annotations**: Encoded across 38 category labels (primary classification target: `related`, denoting emergency relevance).
- **Privacy and License**: Stripped of Personally Identifiable Information (PII) and sensitive data regarding minors. Released under the Creative Commons Attribution 4.0 International License (CC BY 4.0).

Academic Citations

```
@phdthesis{munro12dissertation,
  author       = {Robert Munro},
  title        = {Processing short message communications in low-resource languages},
  school       = {Stanford University},
  year         = 2012,
  address      = {Stanford, CA},
  url          = {https://purl.stanford.edu/cg721hb0673}
}

@article{munro13crisis,
  author    = {Robert Munro},
  title     = {Crowdsourcing and the crisis-affected community - Lessons learned and looking forward from Mission 4636},
```

```
  journal   = {Journal of Information Retrieval},
  volume    = {16},
  number    = {2},
  pages     = {210--266},
  year      = {2013},
  publisher = {Springer},
  url       = {https://robertmonarch.com/research/Mission_4636_Haiti_2010_SMS.pdf}
}

@inproceedings{munro11conll,
  author    = {Robert Munro},
  title     = {Subword and Spatiotemporal Models for Identifying Actionable
Information in Haitian Kreyol},
  booktitle = {Fifteenth Conference on Computational Natural Language Learning
(CoNLL 2011)},
  pages     = {68--77},
  publisher = {Association for Computational Linguistics},
  year      = {2011},
  url       = {https://aclanthology.org/W11-0309/}
}
```

Ethics and Responsible AI Statement

The deployment of Machine Learning models in emergency crisis response carries profound ethical responsibilities. To align with global AI ethics standards (such as GDPR privacy guidelines and human-in-the-loop disaster response frameworks), this project enforces the following ethical principles:

1. **Privacy Preservation and PII Scrubbing**: All communications analyzed in this repository were scrubbed of Personally Identifiable Information (PII), phone numbers, personal names, and precise residential coordinates prior to release. Sensitive messages concerning unaccompanied minors (`child_alone`) were completely excluded to protect vulnerable populations.

2. **Fairness and Mitigation of Demographic Bias**: Disaster response data collected from news media and social networks can contain systemic selection biases (e.g., disproportionate news coverage based on geographic or socioeconomic factors). Our preprocessing pipeline neutralizes dialectal noise and treats message content objectively across direct SMS and news sources.

3. **Human-in-the-Loop Triage Decision Support**: Automated AI predictions in this system are strictly designed as **decision-support indicators for professional human responders**, rather than fully autonomous decision-makers. No emergency assistance is denied automatically by model outputs.

4. **Model Explainability and Auditability**: Through SHAP feature attribution, first responders can verify *why* a message was flagged as high-priority (e.g., due to presence of explicit keywords like `water`, `medical`, or `trapped`), ensuring full algorithmic transparency and accountability.

Project Architecture and Repository Structure

```
d:/4TH_TREMS/ML/CIA3/
├── DATASET/
│   └── disaster_response_messages-main/     # Raw CSV data splits (training,
validation, test)
├── notebooks/
│   └── Disaster_Response_Complete_Assignment.ipynb # Master pre-rendered
presentation notebook
├── src/
│   ├── preprocess.py                        # Data cleaning, text normalization
and TF-IDF pipeline
│   ├── train.py                             # 5-Fold Stratified CV, GridSearchCV
and evaluation
│   └── interpret.py                         # SHAP global importance and summary
beeswarm plots
├── models/
│   ├── tfidf_vectorizer.pkl                 # Fitted TF-IDF vectorizer (10,000 n-
grams)
│   └── best_model.pkl                       # Saved best model (Calibrated Linear
SVM)
├── reports/
│   ├── Presentation_Script_and_Guide.md     # Video script and presentation
timing guide
│   └── figures/                             # 12 saved confusion matrices, ROC
and SHAP PNG charts
├── app/
│   └── app.py                               # Interactive Streamlit Web
Application
└── README.md                                # Formal Project Documentation and
Ethics Statement
```

Reproducibility Instructions

To achieve exact numerical and visual reproduction of all models, evaluation metrics, and SHAP explainability plots, follow these step-by-step instructions.

1. System Requirements and Environment Setup

- Python version 3.9 or higher (Tested on Python 3.13)
- Supported Operating Systems: Windows, Linux, macOS

Install the required library dependencies:

```
pip install scikit-learn pandas numpy matplotlib seaborn xgboost shap wordcloud
streamlit joblib jupyter nbconvert
```

2. Random Seed Configuration

To ensure deterministic reproduction across execution environments, all random state parameters are fixed to `RANDOM_STATE = 42`:

- `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`
- `LogisticRegression(random_state=42)`
- `RandomForestClassifier(random_state=42)`
- `LinearSVC(random_state=42)`
- `XGBClassifier(random_state=42)`
- `shap.maskers.Independent(..., max_samples=100)`

3. Step-by-Step Execution Order

Step 3.1: Data Preprocessing and Cleaning

Execute the preprocessing script to normalize raw text messages and extract train/validation/test splits:

```
python src/preprocess.py
```

Step 3.2: Model Benchmarking and Tuning

Train all 4 classification algorithms using 5-Fold Stratified Cross-Validation and evaluate performance on the independent test dataset:

```
python src/train.py
```

Generated outputs: `models/best_model.pkl`, `models/tfidf_vectorizer.pkl`, and evaluation figures in `reports/figures/`.

Step 3.3: Model Explainability (SHAP Value Calculation)

Generate global feature importance bar charts and summary beeswarm plots:

```
python src/interpret.py
```

Generated outputs: `reports/figures/shap_global_importance.png`, `reports/figures/shap_summary_plot.png`.

Step 3.4: Launch Master Jupyter Notebook

To view or present the master notebook containing all 12 pre-rendered inline plots and tables:

```
cd d:\4TH_TREMS\ML\CIA3
jupyter notebook notebooks/Disaster_Response_Complete_Assignment.ipynb
```

Step 3.5: Launch Interactive Web Dashboard

To run the live interactive web application:

```
streamlit run app/app.py
```

Model Performance and Benchmarking Results

Evaluated on 10,000 sublinear TF-IDF n-gram features on the official test split (2,397 messages):

Algorithm	CV F1-Score	Test Accuracy	Test Precision	Test Recall	Test F1-Score	Test ROC-AUC
Calibrated Linear SVM	**0.7905**	**85.27%**	**0.8476**	**0.8527**	**0.8499**	**0.8753**
XGBoost	0.7517	85.82%	0.8387	0.8582	0.8407	0.8523
Logistic Regression	0.8012	78.14%	0.8727	0.7814	0.8064	0.8838
Random Forest	0.7490	72.92%	0.8502	0.7292	0.7627	0.8382

Key Findings:

1. **Linear Classifiers (Linear SVM and Logistic Regression)** demonstrated superior generalization performance over decision tree ensembles (Random Forest and XGBoost) due to the high-dimensional, sparse text representations generated by TF-IDF vectorization.
2. **Calibrated Linear SVM** achieved the highest overall test F1-score (**0.8499**) and ROC-AUC (**0.8753**), effectively balancing high precision for emergency responders with high recall for disaster victims.

Model Interpretation and SHAP Analysis

Using SHAP (SHapley Additive exPlanations), top global feature contributions influencing disaster relevance predictions were extracted:

1. **High Positive Contributors (Disaster Emergency Indicators)**:
   - Primary crisis nouns: `water`, `food`, `haiti`, `flood`, `earthquake`, `shelter`, `help`, `injured`, `hospitals`, `please`.
   - Primary action verbs: `need`, `trapped`, `died`, `missing`, `send`.
2. **Negative Contributors (Non-Disaster Indicators)**:
   - Conversational and general terms: `news`, `president`, `http`, `thanks`, `good`, `lol`, `game`.
3. **Operational Impact**:
   - Emergency response teams can apply these feature attribution weights to automatically prioritize messages containing critical relief keywords.